

Project Progress Log #2

In this log, you should include **at least** the followings:

- 1. Which data analytic / business intelligence methods do you originally plan to apply? Include diagrams of your modeling plans, e.g., identify clusters then do backward regression, or Agile / Scrum...**

For my dimension, I plan to apply the following methods:

- Data prep (log)
 - applied Ethan's cleaning and transformation code.
- Price grouping
 - grouping the games into four pricing categories based on their original price.
- Correlation Matrix
 - to assess relationships between all transformed input variables, identify multicollinearity, and confirm which variables should be excluded from the regression model.
- Descriptive Statistics
 - computing the count, mean, median, standard deviation, min, and max of revenue per price tier.
- Visualizations:
 - bar chart, boxplot, count chart, and scatter plot to explore revenue patterns across the price tiers.
- One-Way ANOVA
 - to test whether price tier has a statistically significant effect on log-transformed revenue.
- Post-Hoc Tukey Test
 - to identify which specific tier pairs differ significantly.

- 2. Explain why you think those methods would be suitable for your data modeling.**

The reasons these methods are suitable for my data modeling are:

- Data prep (log)
 - revenue in this dataset is positively skewed.
- Price grouping
 - Raw price as a continuous variable is harder to interpret for business stakeholders. Grouping into tiers makes the findings more actionable.

- One-Way ANOVA
 - is suitable for comparing a continuous outcome across more than two groups (log, 4 tiers).
- Post-Hoc Tukey Test
 - after confirming statistical significance with the ANOVA, the Tukey test is applied to identify which specific price tier pairs differ significantly from each other. It also controls for Type 1 error when making multiple pairwise comparisons.

**3. What kinds of tools do you plan to use to implement the model and why?
Different tools have different purposes, please be specific.**

Tools:

- R
 - primary language for all data preparation, statistical testing, and modeling.
- moments
 - computes skewness values before and after transformation to assess and verify which variables needed correction.
- dplyr/tidyverse
 - grouping and summarizing by price tier.
- corrplot
 - it makes it easier to quickly identify strong and weak relationships between transformed variables at a glance.
- ggplot2
 - used for all visualizations (list charts).
- Base R (aov, Tukey)
 - Using One-way ANOVA and Post-Hoc Tukey testing for group comparison.

4. For Data Analytics, you should at least complete some preliminary checking, e.g., matrix plot, please share your results, findings, and thoughts:

- *Which variables are highly correlated with each other?*
 - Excluding log_copiesSold, log_revenue and log_avgPlaytime are the next strongest relationship in the dataset. This suggests that higher-revenue games tend to accumulate more playtime. sqrt_price shows

a weaker correlation to log_revenue, suggesting that price alone is not a dominant predictor of revenue alone. log_reviewScore shows almost no correlation to log_revenue, which suggests that among the top-performing Steam games, review score does not affect revenue that strongly.

- *What else you have done so far?*
 - I have completed the following work for my dimension: applying data prep code, group pricing, correlation matrix, descriptive statistics, one-way anova, and tukey test. I am currently adjusting my visualizations.
- *Any "expected" results? List the expected results and state why they are expected.*
 - I expect Under \$20 to have the highest game count.
 - Most indie and mid-tier games on Steam fall in this price range
 - I expect \$40+ games to have the highest avg revenue per game.
 - AAA title games typically fall in this price range, and have larger marketing budgets and brand recognition
 - I expect \$40+ tier to have higher variance since it only has 65 games.
- *Any "surprising" results? List the surprising results and state why you are surprised.*
 - Free-to-Play and \$20-\$40 are statistically indistinguishable.
 - Free-to-Play games generate comparable revenue to mid-tier range paid titles among the top performers. This tells us that microtransactions and DLC content can be equally as effective as mid-tier range pricing.
 - Under \$20 has the lowest mean log revenue despite accounting for 70% of the dataset.
 - Free-to-Play shows extreme internal variance.
 - It has a mean revenue of \$2.8M against a median of only \$312K. Which shows us that a small number of outliers, such

as CS2 and Dota 2, are heavily skewing the Free-to-Play average upward.

- *Any action plans in mind at this moment?*
 - Note in the limitations section of our report that \$40+ tier only has 65 games, which could limit the reliability of conclusions drawn for that pricing category.
 - Fill out my dimensions report section and mockup a PowerPoint slide.
 - Incorporate sqrt_price as an input factor in the group's Multiple Linear Regression and Random Forest models.

5. For Business Intelligence, you should at least completed some prelim design, please share your:

- Use Case Diagrams

USE CASES

- View revenue dashboard
 - Compare price tiers
- View feature importance report
 - Read insight brief
 - Filter by publisher class

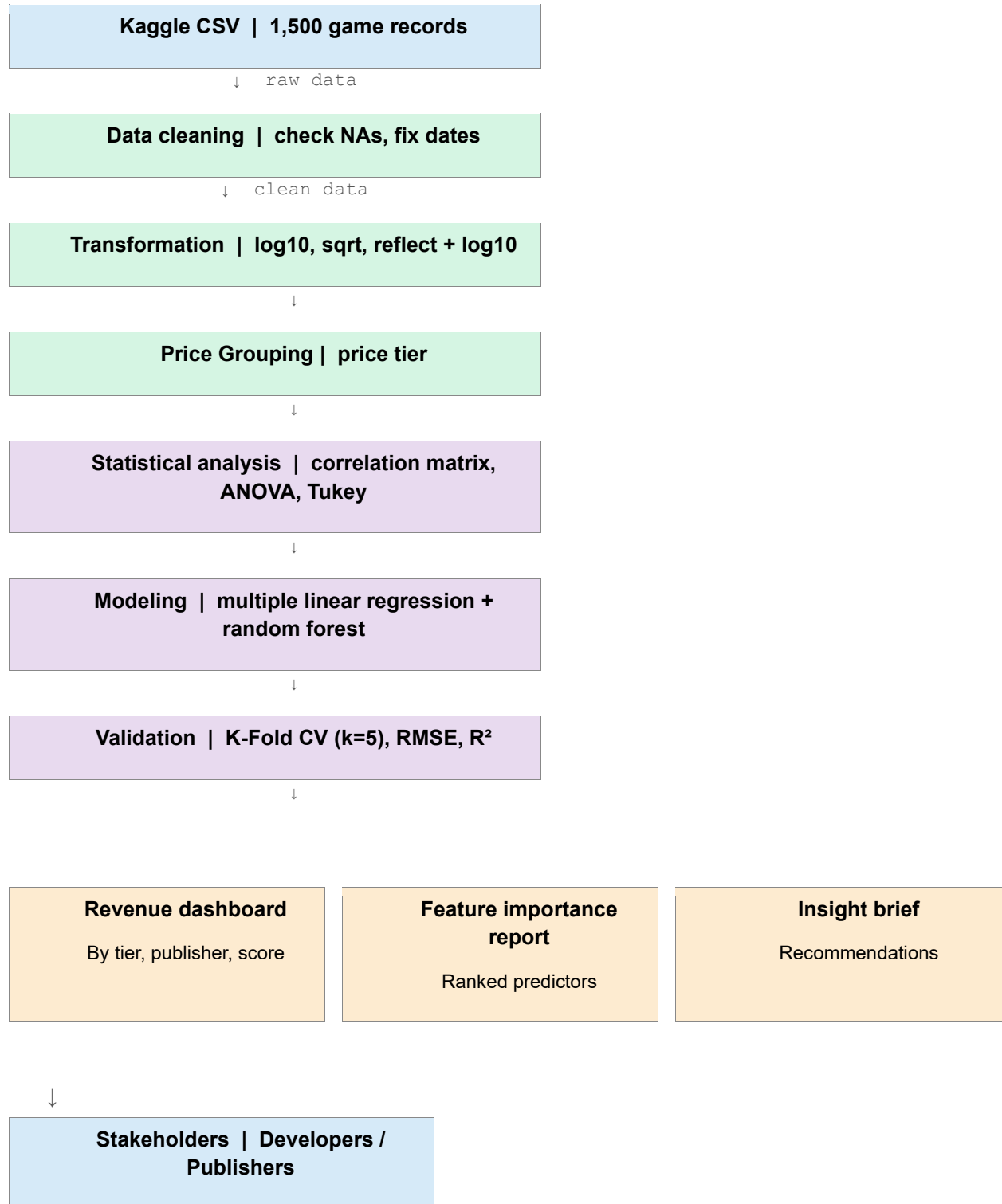
ACCESS

Game Developer → Dashboard, Price tiers, Feature report

Publisher → Insight brief, Publisher filter

Data Analyst → All use cases

- Data Flow Diagram



- Entity-Relationship Diagram

PUBLISHER	GAME	PRICE TIER
🔑 publisherId (PK)	🔑 steamId (PK)	🔑 tierId (PK)
publisherName	name	tierLabel
publisherClass	releaseDate	priceMin
(AAA / AA / Indie)	price	priceMax
1 →	M →	
M	revenue ← TARGET 1	
publishes	avgPlaytime belongs to	
	reviewScore	
	ageOfGame (derived)	
	publisherId (FK)	
	tierId (FK)	

Notes:

- steamId is the primary key uniquely identifying each game record
- publisherClass (AAA / AA / Indie) is derived from the publishers column, used as Dimension 1
- price_tier is engineered from price, used as Dimension 2 (F2P \$0 / under \$20 / \$20–\$40 / \$40+)
- revenue is the target variable (log10 transformed); copiesSold excluded due to circularity (revenue = copiesSold × price)